

HEART DISEASE PREDICTION USING MACHINE LEARNING

Heart Disease Prediction Using Machine Learning

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ABSTRACT

Heart disease is ranked as one of the leading causes of death worldwide. Cases of cardiac diseases such as heart attacks and blood vessel disease are developing at an alarming rate. Therefore, it is critical to foresee any such ailments in advance. However, this is a challenging task that should be done accurately and quickly. Over the years, machine learning (ML) has helped to diagnose various diseases and classify or forecast their outcomes. It is widely known and has been acknowledged to play a significant role in the health sector. This paper employed machine learning algorithms and the Python programming language to predict the possibility of someone having heart disease. Instead of waiting to diagnose patients with heart disease, this system takes a more intuitive approach to predict its likelihood before it occurs so as to aid early treatment, prevention, and a change of lifestyle where necessary. Its main objective is to help individuals know how susceptible they are to heart disease. It used survey data of over 300,000 U.S. citizens collected by the Center for Disease Control (CDC) to build a predictive model. Each feature contains the history of behavioral habits over the period of one year. Three machine learning algorithms, namely Logistic Regression, Random Forests, and Support Vector Machines, were evaluated in this study, of which Logistic Regression had the highest performance of 76% accuracy with an AUC score of 0.84.

Keywords: heart disease, heart disease prediction, machine learning

CHAPTER ONE

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

A variety of illnesses that affect the heart is called heart disease. According to World Health Organization reports, heart diseases are now the leading cause of death worldwide, with 17.9 million fatalities each year. Coronary heart disease, cerebrovascular illness, rheumatic heart disease, and other ailments related to the heart are all types of heart and blood vessel disorders. Heart attacks and strokes are reported to be responsible for more than four out of every five deaths caused by heart disease, with one-third of these deaths occurring in those before the age of 70. Unhealthy food, lack of physical activity, cigarette use, and problematic alcohol consumption are the likely lifestyle risk factors for heart disease and stroke. Individuals may experience symptoms such as hypertension, high blood sugar, increased blood lipids, and obesity as a result of health behaviors. These intermediate-risk factors are detectable in primary care settings and signal a higher chance of a heart attack, stroke, heart failure, and other consequences (WHO, 2019).

As technological know-how increases and technology advances, more patient data can be stored in databases and can be used to carry out healthcare-related research and experiment. There are numerous open sources for gaining access to patient records. With patient data being made easily available, studies can be done with special techniques that can help correctly diagnose patients with the risk of heart disease. Artificial intelligence (AI), which includes machine learning (ML), is widely known and acknowledged to play a significant role in the health sector. It has helped to diagnose various diseases and classify or forecast the outcomes. By using a variety of machine learning algorithms and deep learning models, thorough genetic data analysis can be done, and patient data can be changed and studied more deeply to make better predictions. Also, machine

learning models can be trained to carry out knowledge pandemic predictions (Bharti *et al.*, 2021).

Traditionally, heart disease can be detected by taking blood samples of patients to the laboratory, carrying out cardiac computed tomography scans, cardiac magnetic resonance imaging, electrocardiogram (ECG) and other procedures. The human blood serves as a major indicator of the condition of the heart. Despite this fact, a normal blood test cannot accurately predict the risk of heart disease early. A cholesterol examination and a C-reactive protein (CRP) examination are two standard blood tests for heart disease. These tests check the cholesterol and CRP levels in the blood, and the outcomes can help the doctor have a clear understanding of a patient's overall heart health. An ECG captures electrical impulses, while a heart rate monitor is used as a handheld device that the patient wears for 24 to 72 hours to record a steady ECG. However, traditional diagnostic approaches are time-consuming and intrusive (Shu *et al.*, 2017). On other occasions, doctors can ask their patients a few questions that will help them determine whether they are at risk of developing heart disease or not. This process necessitates a one-on-one appointment with the doctor. Therefore, to ease the challenges faced in the traditional way of predicting heart disease, machine learning algorithms are used to analyse huge, complex medical datasets and create models that can learn based on the data collected, helping medical practitioners predict heart disease.

Machine learning (ML) is a sort of artificial intelligence (AI) that allows computers to solve problems by learning from data. It enables computers to learn and develop independently without having to be expressly programmed to do so. Machine learning systems learn how to or are trained to integrate inputs to make meaningful predictions on unseen data. With an effective machine learning detection algorithm, heart diseases can be predicted with the attributes of patients. Machine learning is a data science methodology that trains computers to learn from experience in the same way that humans and animals do. Machine learning algorithms employ computer approaches to learn knowledge directly from raw data rather than depending on a model based on a preconceived calculation (traditional approach). As the sample data provided for learning grows, the algorithms adjust their performance to respond to the data (Jain, 2019).

Machine learning tasks like classification, clustering, prediction, etc. has proven useful in healthcare. Machine learning techniques can predict how future events will develop based on existing data and common trends. They can also be used to group together comparable medical cases in order to examine patterns and perform future study. Also, regular repetitive chores that requires a lot of time and effort from patients and doctors, such as entry of data, scheduling appointments, inventory control, and so on, can be handled by machine learning. Machine learning, for example, has been shown to be 92 percent accurate in predicting COVID-19 patient mortality. Aside from these, in medicine, machine learning can aid in the development of a more exact treatment plan. Many medical problems are unique and necessitate a particular strategy in order to provide appropriate care and minimise negative effects. The hunt for such solutions can be made easier using machine learning methods. Patient files are massive data sets that require careful analysis and organisation, and machine learning was designed to deal with them. Furthermore, while a health practitioner and a machine learning algorithm will almost certainly reach the same conclusion based on the same data set, machine learning will produce results far more quickly, allowing treatment to begin sooner. Another advantage of employing machine learning techniques in healthcare is that it decreases the likelihood of human error by removing some human participation. This is especially true for process automation activities, as it is in mundane repetitive work that humans make the most mistakes (Tkachenko, 2021).

Heart disease is a life-threatening condition that should not be taken for granted. According to (Havard Health Publishing, 2016), males are more likely than females to develop heart disease. Further research has revealed that men are prone to heart attacks twice as likely as women. Even after taking measures to prevent heart disease risk factors such as the absence of physical activity, hypertension, diabetes, and obesity, the greater risk persisted. This study presents a more effective and accurate way to discover heart diseases quickly by using various medical attributes of the patient. The heart disease prediction system proposed by this study aims to predict those patients with a heart disease or those without a heart disease with machine learning algorithms.

1.2 STATEMENT OF THE PROBLEM

Heart disease is difficult to detect in its early stages, yet early detection is critical for effective treatment. Diagnoses, on the other hand, are made based on medical specialists' personal experiences and perspectives of the condition, which raises the likelihood of errors, inhibits appropriate treatment, lengthens treatment timeframes, and drives up costs. A thorough consideration of these factors has led healthcare bodies like the Centers for Disease Control and prevention (CDC) to gather massive amounts of data from patients and store them in datasets or databases, which are used to build machine learning models.

There are several benefits to switching from traditional methodologies to machine learning. A recent study found that a machine learning model can uncover patterns and forecast outcomes faster than a human or traditional technique. This advantage will enable early treatment of the patient. In addition, the forecasts are endorsed as being highly accurate and constant. Carrying out a diagnosis with machine learning models will reduce human involvement and, by so doing, limit human-related errors. It will also reduce standard costs to an extent.

1.3 AIM AND OBJECTIVES OF THE STUDY

This project aims to build a heart disease prediction system using machine learning. The objectives of this study are as follows:

- I. To carry out extensive research on heart disease prediction with machine learning.
- II. To analyse the need to use machine learning to predict heart disease.
- III. To provide an effective and auxilliary approach for medical specialists and non-medical specialists to carry out the prediction of heart disease.
- IV. To implement a heart disease prediction system that automates and reduces previously manual processes.
- V. To identify gaps in the current way of predicting heart disease and highlight the potential for machine learning to improve previous processes.

1.4 SCOPE OF THE STUDY

This study is limited to developing a heart disease prediction system with machine learning. The system uses a powerful machine learning algorithm as the basis for the model to make predictions. It collects patients' data from the web user interface and feeds them into the machine learning model through a Streamlit application, which serves as the interface between the user and the model.

1.5 SIGNIFICANCE OF THE STUDY

The system proposed by this study will be of great benefit to cardiologists by helping to automate and quicken the process of predicting whether a patient has heart disease or not. Apart from the medical specialists, it will also be handy for citizens to check their likelihood of developing heart disease based on their lifestyle. This method of predicting heart disease is safe, accurate and quick.

1.6 LIMITATION OF THE STUDY

This study is limited by the absence of robust and quality data to develop a heart disease prediction system. Data to fit the requirements of the system could not be gotten from any primary source. This led to the use of secondary data from the World Wide Web.

1.7 DEFINITION OF TECHNICAL TERMS

Clustering: Clustering is the task of dividing the population or data points into a number of groups such that data points in the same groups are more similar to other data points in the same group than those in other groups.

Machine learning model: A machine learning model is a file that has been trained to recognize certain types of patterns.

Optimal transfer learning: the application of knowledge gained from completing one task to help solve a different, but related, problem.

Feature selection: Feature selection is the process of reducing the number of input variables when developing a predictive model.

Regression: Regression is a statistical method used in finance, investing, and other disciplines that attempts to determine the strength and character of the relationship between one dependent variable (usually denoted by Y) and a series of other variables (known as independent variables).

Density estimation: Density estimation is the problem of reconstructing the probability density function using a set of given data points.

Neural network: A neural network is a series of algorithms that endeavors to recognize underlying relationships in a set of data through a process that mimics the way the human brain operates.

Inference: is the process of running data points into a machine learning model to calculate an output such as a single numerical score.

Cross validation: Cross-validation is a resampling method that uses different portions of the data to test and train a model on different iterations.

Undersample: Undersampling is a technique to balance uneven datasets by keeping all of the data in the minority class and decreasing the size of the majority class.

Binary format: A *binary format* is a format in which file information is stored in the form of ones and zeros, or in some other binary (two-state) sequence.

Pickling: Pickle in Python is primarily used in serializing and deserializing a Python object structure.

Hosted: Hosted, past for host means to store (a website or other data) on a server or other computer so that it can be accessed over the internet.

Scripts: A script is a collection of commands in a file designed to be executed like a program.

Classifier: A classifier is the algorithm itself – the rules used by machines to classify data.

Confusion matrix: Confusion matrix is an $N \times N$ matrix used for evaluating the performance of a classification model, where N is the number of target classes.

AUC ROC curve: *AUC-ROC curve* is a performance measurement for the classification problems at various threshold settings.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

A literature review is a survey of works that are relevant to the research that is being conducted. A review of the literature was conducted to see how machine learning can be used to predict the presence of heart disease. Its goal is to uncover and analyze different techniques for predicting heart disease by identifying relevant research publications. As a result, the following information is a critical assessment of previously published works on machine learning-based heart disease prediction.

2.1.1 OVERVIEW OF MACHINE LEARNING

Machine learning refers to a set of techniques that allow a computer to discover relevant patterns and insights from data without requiring human intervention. Using approaches like optimal transfer learning, feature selection, and multitask learning, machine learning can assist a computer in learning more effectively. Machine learning leverages data to perform tasks and performs well with a large amount of data. Human knowledge also contributes to the effectiveness of a machine-learning methodology. Machine learning has been effectively implemented in a wide range of applications and has achieved state-of-the-art growth as a result of this synergy. Furthermore, machine-learning algorithms have lately been used for medical purposes (Fu *et al.*, 2019).

Machine learning is a key discipline in artificial intelligence. It has been used to make brilliant decisions and it can uncover the outcomes of events. The engine of machine learning is its many algorithms. Machine learning tasks can be grouped as supervised, unsupervised, semi-supervised and reinforcement learning. These tasks, when combined with the right algorithm and large volumes of data will increase machine learning effectiveness. Some researchers categorise machine learning based on their classified outputs such as prediction, regression, clustering, and density estimation, all of which rely on algorithms or newer techniques such as artificial neural

networks. These classified outputs refers to the tasks that a machine learning algorithm performs. Naive Bayes, logistic regression and support vector machines (SVM) are examples of supervised algorithms used to perform machine learning tasks. Unsupervised learning, on the other hand categorises ambiguous data using extracted characteristics and comparison (Al-Khasawneh *et al.*, 2022).

Machine learning systems when correctly implemented, may remove human participation partially or entirely, leading to fewer mistakes. Also, automating and speeding up previous manual processes, making certain daily tasks more comfortable for citizens and even lower long-term standard costs for some procedures. While there are some machine learning systems already actualising a few of these advantages, they still fall short after considering some criteria.

In 1950, Alan Turing developed the Turing test to determine whether a computer has human-like intelligence. In 1952, Author Samuels wrote the first machine learning program. 1957 experienced the design of the first neural network by Frank Rosenblatt. It implemented the human brain's thinking process and was called the perceptron. In the 1990's machine learning methodology moved from knowledge-driven process to data-driven. The 2010s saw the development of powerful machine learning machines and new algorithms. This timeline shows how far machine learning has come, and its exploits (Kot, 2021).

2.1.2 CHARACTERISTICS OF MACHINE LEARNING

According to Ray (2019) machine learning is about computer programs performing tasks and learning from its experience. The following are characteristics that a machine learning program exhibits:

- I. Automation: Machine learning automates repetitive human tasks.
- II. Learning from experience: It gains more experience as it executes tasks and as more data is given to it.
- III. Intelligence: Ability to think and make accurate predictions from data.
- IV. Accuracy: Gives correct and dependable predictions.

- V. Scalability: An increase in amount of data does not decrease performance and there is no wastage of memory.
- VI. Speed: A machine learning program can predict outcomes in seconds.
- VII. Autonomous: Machine learning programs independently forecast outcomes without any human intervention.

2.1.3 IMPETUS FOR MACHINE LEARNING

A typical traditional piece of software requires the programmer to create rules for each of the tasks the software is supposed to perform. For every pattern the programmer observes, he writes a new rule for it. This will lead to the need to write a new set of rules anytime a new pattern is discovered. In contrast, machine learning software will automatically learn patterns and generalise well when it comes across new patterns. This makes the program shorter, easier to maintain and more accurate. Another driving force for machine learning is that it is perfect for problems that do not have a known algorithm, solutions that are fluctuating in nature, or tasks that are too complex for the traditional approach.

Machine learning is a substitute for the traditional prediction modelling approach, which entails using a computer to learn about complex and nonlinear interactions between several variables by reducing the difference between expected and actual results.

Machine learning discovers patterns that were not seen before and helps humans learn from the patterns that were discovered.

2.1.4 PROBLEMS OF MACHINE LEARNING

A major challenge of machine learning is the lack of quality data and insufficient quantity of data to carry out tasks. When there is lack of quality data, the models created may end up with a large amount of inaccurate predictions. According to (Geron, 2017), machine learning is not there yet; most of its algorithms require a lot of data to function properly. Thousands of instances are usually required for simple problems, while millions of examples may be required for complicated problems such as image or speech recognition. Another issue is the use of a bad algorithm. Not all algorithms are suitable for all types of data. A defective algorithm can either

overfit the training data, causing the model to fail to generalize, or underfit the training data, causing the model to fail to understand the hidden patterns in the data used to train it, resulting in inaccuracy.

These challenges can be solved by creating a model with the right algorithm and training more skilled machine learning engineers.

2.1.5 SUPERVISED LEARNING

Most machine learning applications use supervised learning. In supervised learning, the training data fed to the algorithm chosen includes the desired answer or solution. A model that wants to predict outcomes is fed with data called the training data which contains the expected result from the prediction. This heart disease prediction system uses supervised learning. The labels or solution is given, that is 'Yes' for heart disease and 'No' for no heart disease.

According to (Brownlee, 2020), supervised learning involves a scenario where you have input values (x) and a predicted value (Y), and then using an algorithm to understand the mapping function from the input to the output. A mathematical equation for supervised learning is given as $Y = f(X)$. The objective is to estimate the mapping function to the point that you can forecast the output values (Y) for new input data (x). The act of an algorithm learning from the training data is referred to as supervised learning since the learning process is like a student learning from a teacher, in this case, the algorithm learns from the training data. The algorithm iteratively produces forecasts on the training instances and is corrected by the teacher (training data). When the algorithm reaches a satisfactory level of performance, learning comes to an end.

There are two types of supervised learning problems, namely classification and regression. A classification problem classifies the predicted outcome as a category, such as "Yes" or "No," or "disease" or "no disease". A regression problem forecasts the output value as an integer or number, such as "naira" or "weight".

The system discussed in this study is a classification problem because it forecasts the output variable as "Yes" and "No" replaced with a "1" and "0" respectively, and then, with some basic

mathematical calculation, converts it into a percentage value. The supervised learning algorithm employed by this study is logistic regression. It is frequently used to calculate the likelihood that an instance belongs to a specific class. If the calculated probability is more than 50%, the model predicts that the instance belongs to that class, that is, the positive class with the label "1", or that it does not, that is, the negative class with the label "0". As a result, it is a binary classifier. More about this algorithm will be discussed later in this study.

2.1.6 LOGISTIC REGRESSION

Logistic Regression (also called Logit Regression) is commonly used to estimate the probability that an instance belongs to a particular class (e.g., what is the probability that this email is spam?). If the estimated probability is greater than 50%, then the model predicts that the instance belongs to that class (called the positive class, labeled "1"), or else it predicts that it does not (i.e., it belongs to the negative class, labeled "0"). This makes it a binary classifier (Geron, 2017).

Just like a Linear Regression model, a Logistic Regression model computes a weighted sum of the input features (plus a bias term), but instead of outputting the result directly like the Linear Regression model does, it outputs the logistic of this result.

$$p = h_{\theta}(x) = \sigma(\theta^T \cdot x)$$

Logistic Regression model estimated probability (vectorized form)

The logistic—also called the logit, noted $\sigma(\cdot)$ —is a sigmoid function (i.e., S-shaped) that outputs a number between 0 and 1.

Once the Logistic Regression model has estimated the probability $p = h_{\theta}(x)$ that an instance x belongs to the positive class, it can make its prediction $\hat{y}$ easily.

2.2 CONCEPTUAL FRAMEWORK

Machine learning is concerned with the creation of programs that allow machines to learn through making inferences from observation data that represents partial information about statistical phenomena. Classification, also known as pattern recognition, is a crucial form of machine learning in which computers "learn" to recognize complicated patterns automatically, differentiate between samples based on their patterns, and make intelligent judgments. The data representation (feature extraction) module, feature selection or reduction module, and classification module are the three modules that make up a pattern classification task. The first module's goal is to uncover uniform features that can best explain class differences. The second module's goal is to reduce the feature vectors' dimensionality for classification. The classification module uses features to determine the actual mapping between patterns and labels (Cen *et al.*, 2010).

Acquisition of a large amount of experimental data is crucial for machine learning (i.e., the total sample size should be greater than the number of parameters to be calculated). The greater the number of samples, the better. Part of the data (training set) is used to build a model (program), after which the model is tested on the remaining data (test set). This strategy will help reduce the risk of overfitting—that is, when a model predicts better on seen data but worse on unseen data—and so ensure the constructed model's trustworthiness even when new experimental data becomes available. Machine learning uses cross-validation in addition to the train-test split technique to help prevent overfitting and choose hyper-parameters (parameters used in the learning process). The training set is broken into numerous subsets when the cross-validation technique is used. The validation set is used to assess the developed model's prediction power, while the rest is used to build the model. The model construction and validation process is repeated until each sub-fold is used as a validation set. The mean of each performance value in the loop is used to calculate the model's final fitness after using the cross-validation procedure (Zhang and Wu, 2021).

Machine learning algorithms that are successful can perform a variety of tasks. A machine learning system's function can be descriptive, which means it uses data to describe what happened; predictive, which means it uses data to anticipate the event that will occur; or prescriptive, which means it uses data to provide recommendations on what action to take (Brown, 2021). The goal of machine learning is to gain knowledge from data. A training and testing dataset is often created from the input data. The output in the training dataset has to be predicted or categorized. For prediction or classification, all algorithms learn patterns from the training sample and use them on the test dataset (Mahesh, 2019).

2.3 THEORETICAL FRAMEWORK

The theoretical background gives an overview of the general concept of the research topic as seen by other researchers. Machine learning creates a platform for automating tasks in general. Specifically, in healthcare, machine learning is used to analyze millions of distinct data sources, predict outcomes, and provide fast risk scores and exact resource allocation, among other things. Machine learning assists practitioners in gaining access to more data so that they may make better decisions regarding patient diagnoses and possible treatments while also comprehending the potential consequences and costs associated with each. The benefit of machine learning in healthcare is its capacity to process large datasets beyond human competence and then accurately transform that analysis into clinical insights that assist doctors in planning and giving care, eventually leading to better results, cheaper healthcare costs, and increased patient satisfaction.

Purushottam *et al.*, (2016) introduced a heart disease prediction system that uses decision trees and hill climbing algorithms to predict heart disease. They used the Cleveland dataset, which was preprocessed before classification techniques were applied. The Knowledge Extraction is based on Evolutionary Learning (KEEL), an opensource data mining method that fills in the data set's missing values. The order of the decision tree used is top-down. A node is chosen by a test at each level for each actual node chosen by the hill-climbing algorithm. The system has an 86.7 percent accuracy rate.

Beyene and Kamat (2018) in a study on the prediction and analysis of the occurrence of heart disease with data mining techniques developed a prediction system that forecasts the emergence of cardiac disease so that the condition can be diagnosed automatically and quickly. The methodology proposed by this study is also important in a healthcare organization where not all the doctors have the same knowledge and skill to diagnose heart disease; they make their own decisions, which can lead to poor outcomes and even death. It analyses many physiological characteristics like heart rate and blood sugar, as well as age and gender to determine whether or not a person has heart disease. WEKA software was used to compute dataset analysis.

Devari and Khadem (2016) developed a system to automate diagnosis of coronary artery disease with optimised Support Vector Machines (SVM). The study showed how to use optimised SVM to diagnose individuals with heart disease. In this work, SVM parameters were modified to improve prediction accuracy, resulting in a 99.2 percent accuracy using k-fold cross-validation. The solution presented in this paper aids in the early detection of heart disease and lowers the expense of regular tests. The accuracy acquired is sufficient to determine whether or not the individual has heart disease.

Golande and Kumar (2019) examined some machine learning algorithms for classifying heart disease. The accuracy of decision trees, k-nearest neighbors algorithms, and K-Means algorithms that could be used for classification were investigated. This research shows that decision trees have the best accuracy and can be made more effective by combining different methods and changing parameters.

Khemphila and Boonjing (2011) proposed a cardiac condition detection system that uses a classification method based on neural networks and feature selection, introducing a multi-layered perception and backpropagation algorithm. The neural networks were used to classify patients with heart disease or not. It was used because it increases classification accuracy. These predictions achieved an accuracy of 89.56% on the training data set and 80.99% on the test set.

Jeyaganesan *et al.*, (2020) used K Nearest Neighbours (KNN), Logistic Regression and Random Forest Classifiers to develop a system to help practitioners and medical analyst accurately diagnose heart disease. Their study focused on determining the best machine learning algorithm for predicting heart disease. From their research, KNN and Logistic Regression obtained the highest accuracy while Random Forests Classifier produced a better result.

Mohan *et al.*, (2019) proposed a novel method that aims at finding significant features by applying machine learning techniques resulting in improving the accuracy in the prediction of cardiovascular disease. The prediction model is introduced with different combinations of features and several known classification techniques. They produced an enhanced performance level with an accuracy level of 88.7% through the prediction model for heart disease with the hybrid random forest with a linear model (HRFLM).

Chen *et al.*, (2011) developed a heart disease predict system that can assist medical professionals in predicting heart disease status based on the clinical data of patients. Their approaches include three steps. Firstly, They selected 13 important clinical features, i.e., age, sex, chest pain type, trestbps, cholesterol, fasting blood sugar, resting ecg, max heart rate, exercise induced angina, old peak, slope, number of vessels colored, and thal. Secondly, they developed an artificial neural network algorithm for classifying heart disease based on these clinical features. The accuracy of prediction is near 80%. Finally, they develop a user-friendly heart disease predict system (HDPS). The HDPS system consisted of multiple features, including input clinical data section, ROC curve display section, and prediction performance display section (execute time, accuracy, sensitivity, specificity, and predict result).

Parthiban and Subramanian (2008) developed a new approach based on coactive neuro-fuzzy inference system (CANFIS) was presented for prediction of heart disease. The proposed CANFIS model combined the neural network adaptive capabilities and the fuzzy logic qualitative approach which is then integrated with genetic algorithm to diagnose the presence of the disease. The performances of the CANFIS model were evaluated in terms of training

performances and classification accuracies and the results showed that the proposed CANFIS model has great potential in predicting the heart disease.

Khan (2020) proposed an IoT framework to evaluate heart disease more accurately using a Modified Deep Convolutional Neural Network (MDCNN). The smartwatch and heart monitor device that is attached to the patient monitors the blood pressure and electrocardiogram (ECG). The MDCNN is utilized for classifying the received sensor data into normal and abnormal. The performance of the system is analyzed by comparing the proposed MDCNN with existing deep learning neural networks and logistic regression. The results demonstrate that the proposed MDCNN based heart disease prediction system performs better than other methods. The proposed method shows that for the maximum number of records, the MDCNN achieves an accuracy of 98.2 which is better than existing classifiers.

2.4 SUMMARY

Due to the growing popularity of machine learning, multiple heart disease prediction systems are being developed to automate to an extent previous manual processes. Various machine learning algorithms have been applied by researchers to predict heart disease. ANN has been found to perform better in terms of giving the highest prediction accuracy. Other classification algorithms such as SVM, logistic regression, decision trees, KNN, and k-means have also proved to be very effective in predicting heart disease. Studies conducted by researchers show that machine learning algorithms are able to predict not just heart disease but any type of disease now and in the future.

CHAPTER THREE

SYSTEM ANALYSIS AND DESIGN

3.1 INTRODUCTION

This chapter presents the system architecture and research methodology adopted and also analyses the existing methods. The research methodology includes data collection and techniques adopted for developing the proposed system. This chapter will define the approach to achieving the development of the system while emphasizing the problem statement and the system details such as system architectural design, user interfaces, and data processing.

3.2 RESEARCH METHODOLOGY

The methodology contains an overview of the research design, types of data to be collected, sample design, and proper interpretation for the specific research, as well as the required instruments for the research's hypotheses.

Heart disease has lately become one of the world's leading deadly diseases. As a result, we require accurate, rapid, and intelligent technologies for heart disease prediction. Machine learning algorithms were used in this study to diagnose heart disease early and give relevant recommendations. The research of this study is solely limited to the internet. Different research papers by other researchers were consulted.

The system is an end-to-end heart disease prediction web application. It receives data from patients and processes the data with three different machine learning algorithms. The highest performing model was selected and used to implement the system in question. The output after classification is the probability of the patient having heart disease.

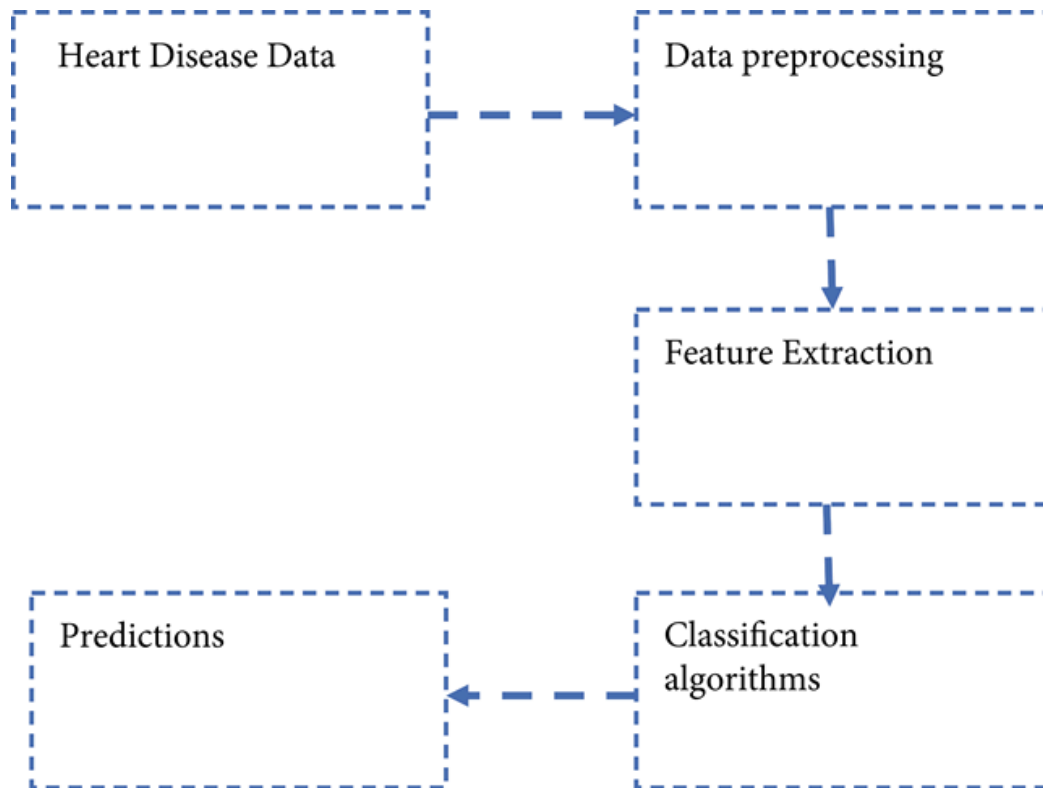


Figure 3.1: Methodology of the Proposed System

Source: (Phasinam *et al.*, 2022)

Figure 3.1 the methodological approach the system will take starting with data gathering to predicting heart disease.

3.2.1 Data Collection Methods

During the building and research of this project work, data collection was limited to the internet. Due to the unavailability of primary data, secondary data was used. The dataset used in this study is an annual telephone behavioral risk factor survey of the health of United States residents from the year 2020, provided by the CDC. The data was originally stored in SAS format. The original dataset contains about 402,000 rows and over 200 features. The data was then downloaded and data cleaning and conversion were performed to extract necessary features and clean out unnecessary rows. Also, the dataset was heavily imbalanced. I had to undersample the original

data at the time of training the model. The machine learning model was trained on approximately 27,400 rows and 17 features of the original data.

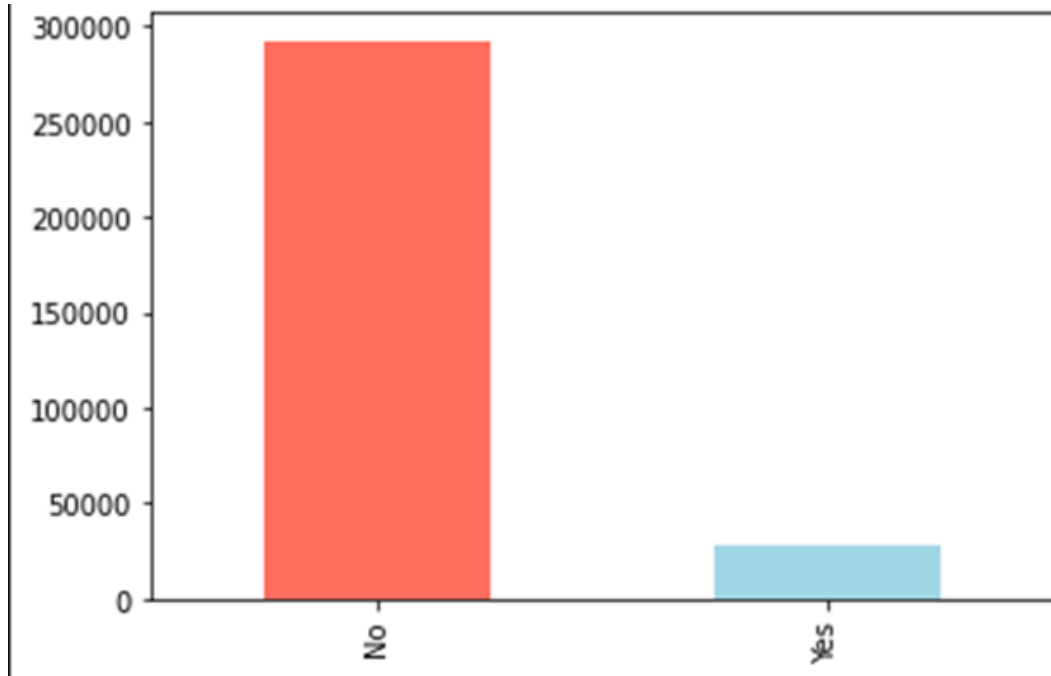


Figure 3.2: Initial Distribution of Data

The heavy imbalance on the data is shown in figure 3.2 above.

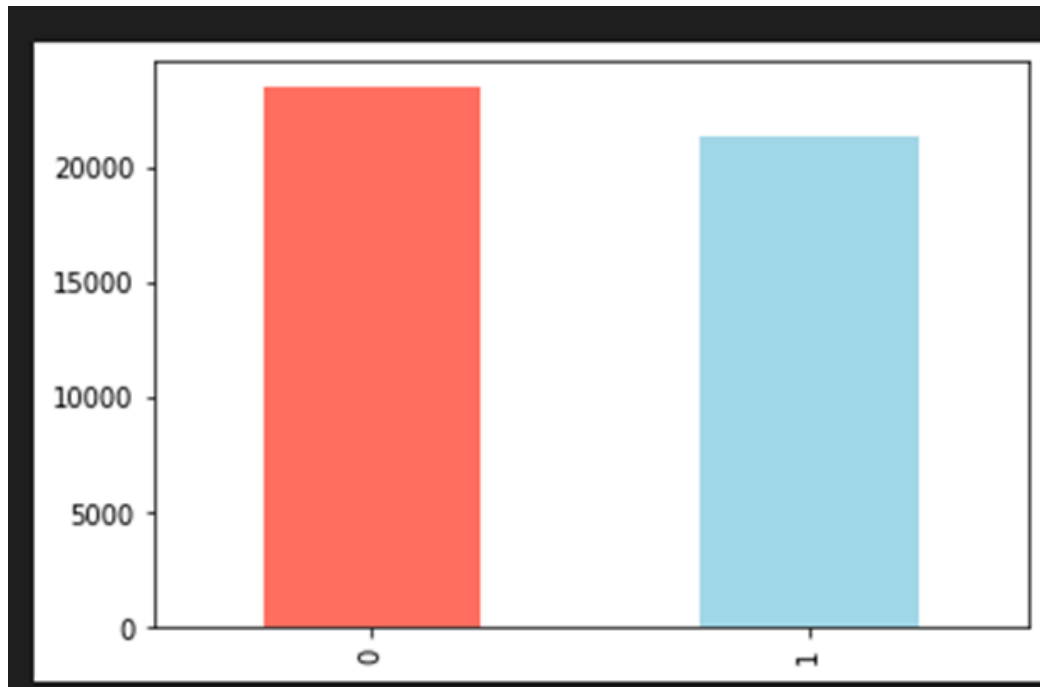


Figure 3.3: Undersampled Data Used for the System

The undersampled distribution is shown in figure 3.3 above. The data was undersampled to balance the class distribution.

3.3 SYSTEM ANALYSIS

System analysis is a problem-solving technique that involves understanding the process involved, identifying problems, and recommending feasible suggestions for improving the system's functioning. This entails researching the business process, obtaining operational data, comprehending the information flow, identifying bottlenecks, and developing a solution to overcome system shortcomings to accomplish corporate goals.

3.3.1 Analysis of the Case Study

Heart disease diagnosis in Nigeria and a host of other nations is achieved by taking blood samples of patients to the laboratory. This process initially begins with the patient noticing an abnormality in his body system. The patient then goes to a doctor or books an appointment with a doctor to report the symptoms he observed in his body. The doctor then collects blood samples

from the patient and sends them to a laboratory to determine the medical challenge of the patient. After some hours or days, the result from the laboratory is sent back to the doctor, who then diagnoses if the medical challenge the patient has is heart disease or not. If the condition is heart disease, the doctor proceeds to commence treatment of the patient. If not, the doctor can refer the patient back to the laboratory again to undertake another test.

Despite the tedious and prolonged procedure, this approach is still in use in the majority of African countries today. Due to the high expense of development and implementation, several countries find it difficult to adopt any new approach for diagnosing heart disease and instead focus on enhancing the current process.

3.3.2 Weaknesses of the Present System

The conventional heart disease diagnosis system has been found to have the following weaknesses that are enumerated below:

- I. Delay: The time between a patient's visit to the doctor and the time the result is released usually spans several days. Heart disease needs to be detected quickly so that early treatment can begin.
- II. Cost: The cost of running tests, consultation, and logistics takes significant financial resources from a patient. The less privileged are unable to fund this type of care.
- III. Tedious: Detecting heart disease manually takes a lot of time and demands more work hours from medical specialists than usual.
- IV. Inefficient: Doctors can make mistakes and take a symptom that is not heart disease for heart disease causing patients to undertake treatment for the wrong sickness.

3.3.3 System Justification

Machine learning proves to be the approach that fits the bill for rapid diagnosis, reduced expenses, and efficiency. Patients input their parameters into the system, which is then processed and fed into a model that uses a machine learning algorithm (Logistic Regression) to analyse the

data of the patient. The trained model learns and uncovers hidden patterns in the data received and then predicts whether the patient has heart disease or not.

3.4 ANALYSIS OF THE PROPOSED SYSTEM

Considering the limitations and challenges experienced by the existing system, there is a need to design an automated and rapid heart disease prediction system with machine learning. The system is very flexible and easy to use. It has features that ensure prediction is fast and accurate.

The proposed system designed in this study is web-based and is open to both doctors' and patients' usage.

New data, i.e. the patient's data, is fed into the predictive model, as shown in Figure 1. The model then predicts whether the patient has heart disease or not.

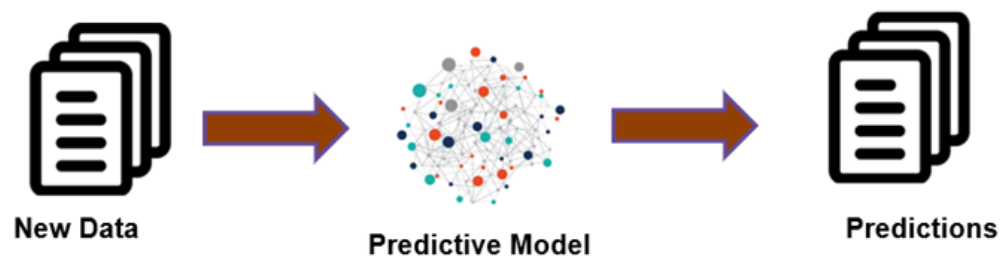


Figure 3.4: Movement of Patient Data in Machine Learning Pipeline

Source: Milleplateaux (2022)

Figure 3.4 shows the movement of data in the machine learning pipeline. An ML pipeline refers to the processes involved before the final prediction system is implemented.

3.4.1 Input Analysis

The dataset used in this study has 17 parameters that will be presented to the user. The model analyzes these data inputs, and the result is a conceptual description of the system under

investigation. The development of the system will be based on this informal description. It is crucial to correctly analyze the collected data.

Parameters

Select your gender

Male

Enter BMI

0.00

For how many days during the past 30 days was your physical health not good?

0.00

For how many days during the past 30 days was your mental health not good?

0.00

How many hours on average do you sleep?

0.00

Have you smoked at least 100 cigarettes in your entire life (approx. 5 packs)?

No

No

Have you played any sports (running, biking, etc.) in the past month?

No

How can you define your general health?

Excellent

Do you have asthma?

No

Do you have kidney disease?

No

Do you have skin cancer?

No

Predict

Figure 3.5: Input Interface Screen

3.4.2 Process Analysis

The data fed into the model passes through a data processing stage before being evaluated. Machine learning algorithms cannot make predictions on strings or objects. They can only make predictions on numerical data. Therefore, data cleaning must first be performed before passing the data to a model. The cleaned data is then fed into the model. Before any prediction is made, the data is split into a training set (80%) and a testing set (20%). The training set is used to train the model with Logistic Regression after which the now trained model is validated with the testing set. After the validation, the accuracy score and the classification report are evaluated.

The model is then saved in a binary format (pickling) so that it does not need to be trained every time the application is ran.

The pickled model is loaded into a Streamlit application with the help of pandas library to display it to the users.

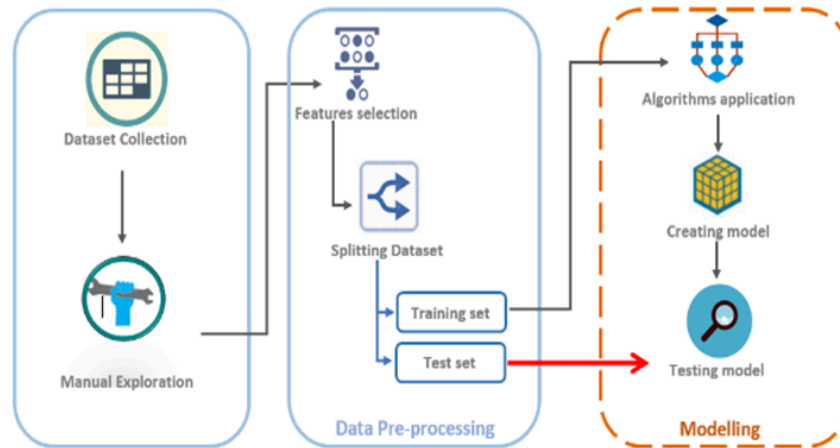


Figure 3.6: Data Processing Flow

Source: Salhi *et al.*, (2021)

Figure 3.6 shows the flow of data in an abstracted view. The data is first collected and then data analysis is done, and features are selected before it is split and trained to make predictions.

3.4.3 Output Analysis

A computer system or mobile device output is generally used to communicate the process's result to the users. After filling in the parameters the model is expecting on the web interface, the user clicks the “predict” button and the application shows the probability of the patient having heart disease in the form of a percentage.

Do you have skin cancer?

Yes

Do you have skin cancer?

No

Predict

The probability that you'll have a heart disease is 91.91%. You need to see a doctor.

Do you have skin cancer?

No

Do you have skin cancer?

No

Predict

The probability that you'll have a heart disease is 0.84%. You are healthy.

Figure 3.7: Output Interface

The output interface above shows the prediction results from the system.

3.4.4 Benefits of the Proposed System

The proposed system has been found to have a number of advantages over the existing system, some of the advantages are:

- I. Speed: Machine learning speeds up the process of initial heart disease prediction or diagnoses. It replaces to an extent the need for patients to go to a laboratory and perform tests thereby eliminating the need to wait for days.

- II. Cost effective: Since machine learning bridges the gap in the traditional process by eliminating the need for patients to carry out tests first before heart disease can be diagnosed, it helps to save the financial resources of the patient.
- III. Efficiency: With machine learning, the tedious work of the medical practitioners will be automated thereby increasing productivity.

3.5 SYSTEM DESIGN

The proposed system was designed using Streamlit and Python. The development of this system is divided into two phases namely, frontend and backend.

The frontend consists of the following framework:

- I. Streamlit

The backend part of this program was developed using:

- I. Python

The details of how this model was arrived at and the training steps are discussed below.

The system proposed in this study is a classification problem under machine learning. Classification problems are a broad aspect of supervised learning. Supervised learning, as discussed in chapter 2.1.5 means training a model to carry out a task with the labels or answers to the problem it wants to solve. Building a machine learning model involves seven major steps referred to the universal machine learning workflow which were employed in this study for building and training the model. The steps are discussed below:

1. Gathering data

The goal of the proposed system is to build a model that can predict the likelihood of a person having heart disease. The first step is to get the data that will be used to build the predictive model. Data is the life of machine learning and without data, machines cannot learn. This step is very important because the quality and quantity of data that you gather will directly determine how good your predictive model can be. In this case, the data that

will be collected will be serveral behavioral information of people. The data has 319,795 rows and 17 features.

BMI	Smoking	AlcoholDrinking	Stroke	PhysicalHealth	MentalHealth	DiffWalking	Sex	AgeCategory	Race	Diabetic	PhysicalActivity	GenHealth
16.60	Yes	No	No	3.0	30.0	No	Female	55-59	White	Yes	Yes	Very good
20.34	No	No	Yes	0.0	0.0	No	Female	80 or older	White	No	Yes	Very good
26.58	Yes	No	No	20.0	30.0	No	Male	65-69	White	Yes	Yes	Fair
24.21	No	No	No	0.0	0.0	No	Female	75-79	White	No	No	Good
23.71	No	No	No	28.0	0.0	Yes	Female	40-44	White	No	Yes	Very good

Figure 3.8: Sample of Behavioral Training Data Used

2. Data preparation

Data preparation involves loading the data into a suitable place and prepare it for use in the machine learning training. First, the data was put into a Pandas dataframe to perfrom futher analysis on the data. Pandas is a python library widely used in data science for data analysis. Exploratory Data Analysis (EDA) was carried out to understand the data in depth and learn the different data characteristics with visual metrics and Pandas functions. From the data analysis, it was discovered that the data contains 4 categorical attributes, 4 numerical attributes and 9 boolean attributes. No null columns were found and the data types of the attributes were found to be object and float64. The label / answer was found to be heavily imbalanced at the time of data visualization. **Figure 3.2** above shows the data imbalance.

All the boolean and categorical columns were replaced with numerical values. This is beacuse the machine learning algorithm can only work with numerical values. A plot for correlations was also done with a Seaborn heatmap to find any correlation in the dataset. From the plot for correlations it was found that the label or target variable is slightly positively correlated with the 4 other numerical variables.

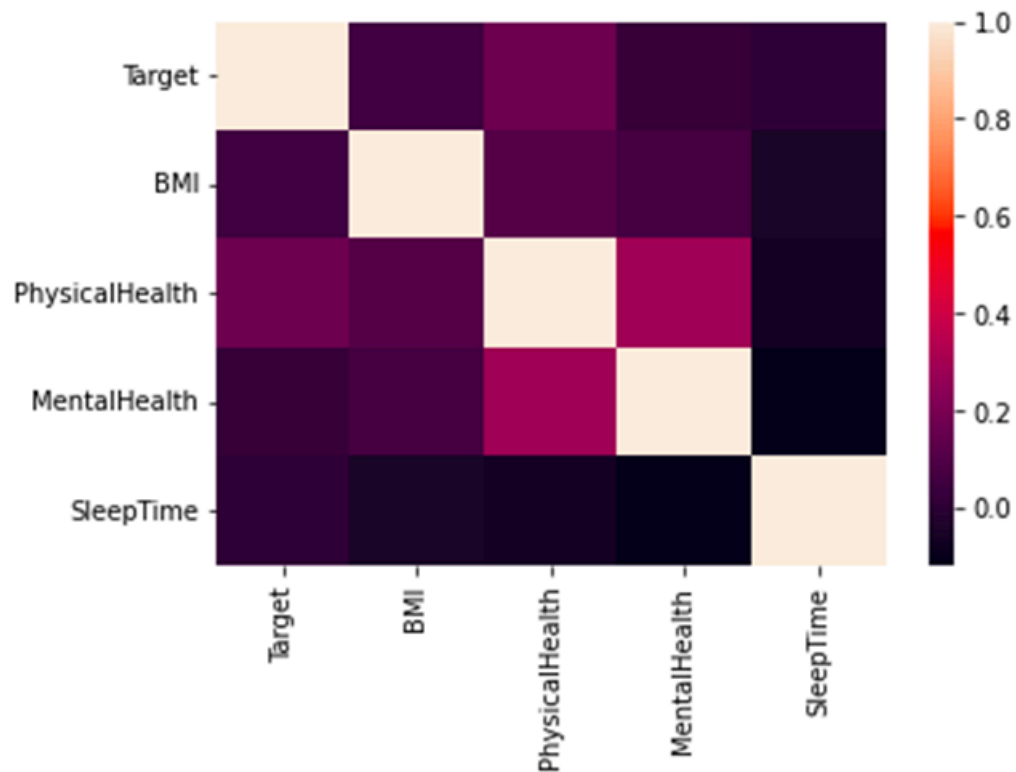


Figure 3.9: Correlation Plot for Data

After the EDA, the data was split into two parts. The first part, used in training the model will be the majority of the dataset (80%). The second part will be used for evaluating the trained model's performance (20%). It is advisable to use a separate data that the model has not come across to test the model.

1. Choosing a model

The next step in the machine learning workflow is to choose a model. There are many models that researchers and data scientists have created over the years. Some are very well suited for image data, others for sequences (like text, or music), some for numerical data, others for text-based data. In the case of the proposed system, since the task we want it to carry out is a basic 'heart disease' and 'no heart disease' prediction, a logistic regression model was used. Its main purpose is to estimate the probability of an event

occurring. The outcome of logistic regression is a probability, the dependent variable is bounded between 0 and 1. Therefore, it is suitable to get the job done.

2. Training

In this step, the prepared data is used to incrementally improve the model's ability to predict whether a person has heart disease or not. The model starts learning gradually and incrementally from the data just like how we learn how to read. At first, it does not know which person has heart disease or which person doesn't. After lots of training iterations with logistic regression, the model becomes quite adept. The act of training and learning from the data has adapted its prediction abilities.

The training process involves initializing some random values for the weight and biases and attempting to predict the output with those values.

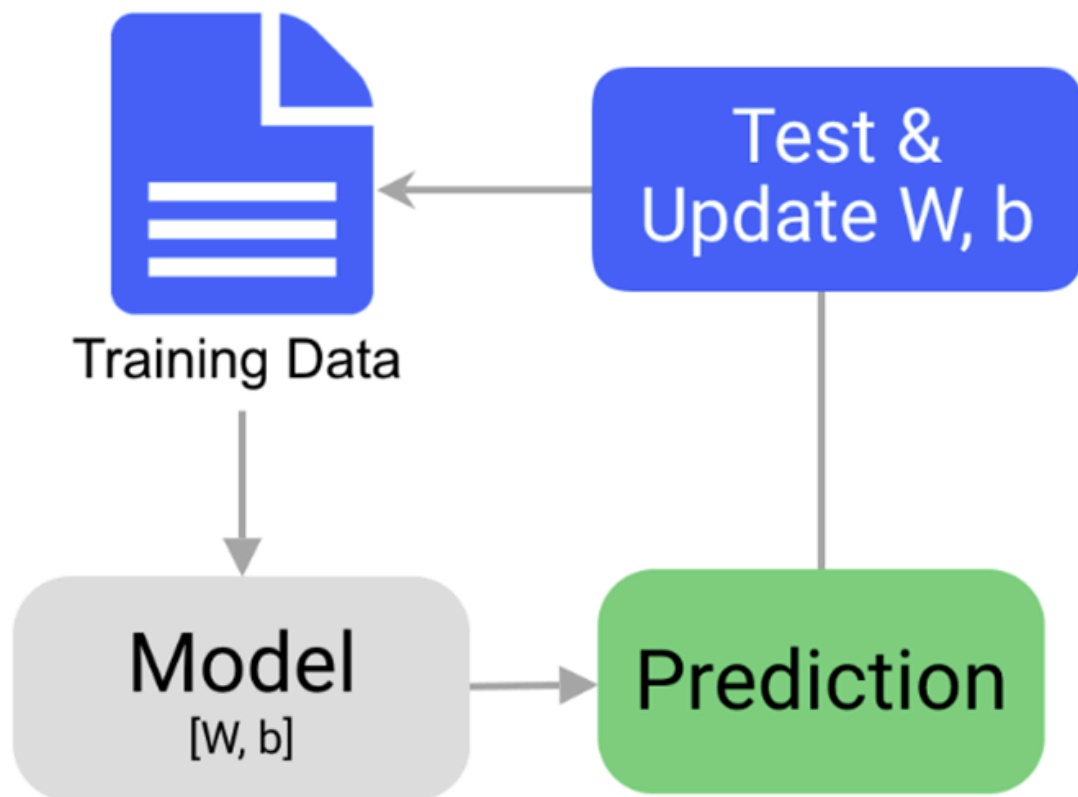


Figure 3.10: Steps Taken in Model Training

This process then repeats. Each iteration or cycle of updating the weights and biases is called one training “step”.

1. Evaluation

After the training was complete, evaluation was done to see if the model performed according to the aim of the study. This is where the 20% of the dataset that was set aside comes into play. Evaluation allows the model to be tested against data that has never been used for training. This metric allows shows how the model might perform against data that it has not yet seen. This is meant to be representative of how the model might perform in the real world.

2. Parameter Tuning

After evaluation, the model was found to have an accuracy of 91%, that means, it was able to predict the probability of a person having heart disease with an accuracy of 91% out of 100%. But evaluating a models’ performance with only the accuracy can be very deceiving. In the case of this system, the model gave an accuracy of 91% but it was not able to correctly classify heart disease or no heart disease because of the heavy imbalance mentioned in number 2 of this workflow. To counter this setback, some major parameter tuning was done to undersample the majority class from a 90:10 sample to 50:50 sample. Before, the ‘no heart disease’ class had the higher ratio therefore the model always predicted that the person will not have heart disease. But after after using the `class_weight` parameter of logistic regression to balance the classes and employing imblearns’ `RandomUnderSampler`, the classes became balanced as shown in **Figure 3.3** above. After the second evaluation was done with the percision, recall, f1-score and confusion matrix, the model performed well and was able to predict ‘heart disease’ and ‘no heart disease’ with a reduced accuracy of 75% but excellent performance on prediction.

3. Prediction

The layman meaning of machine learning is using data to answer questions. Prediction, or inference, is the step where we get to answer some questions. This is the point of all this work, where the value of machine learning is realized. The model can finally be used to predict the likelihood of a person having heart disease or not.

3.5.1 Architecture Diagram of the Proposed System

Architectural design entails understanding how the system should be organised and the appropriate development of the system's general structure. It serves as a vital link between design and requirements engineering by identifying the most important structural components of a system and their relationship, as well as providing an overview of the new system, including the most crucial parts of the system, the services they provide, and their communication.

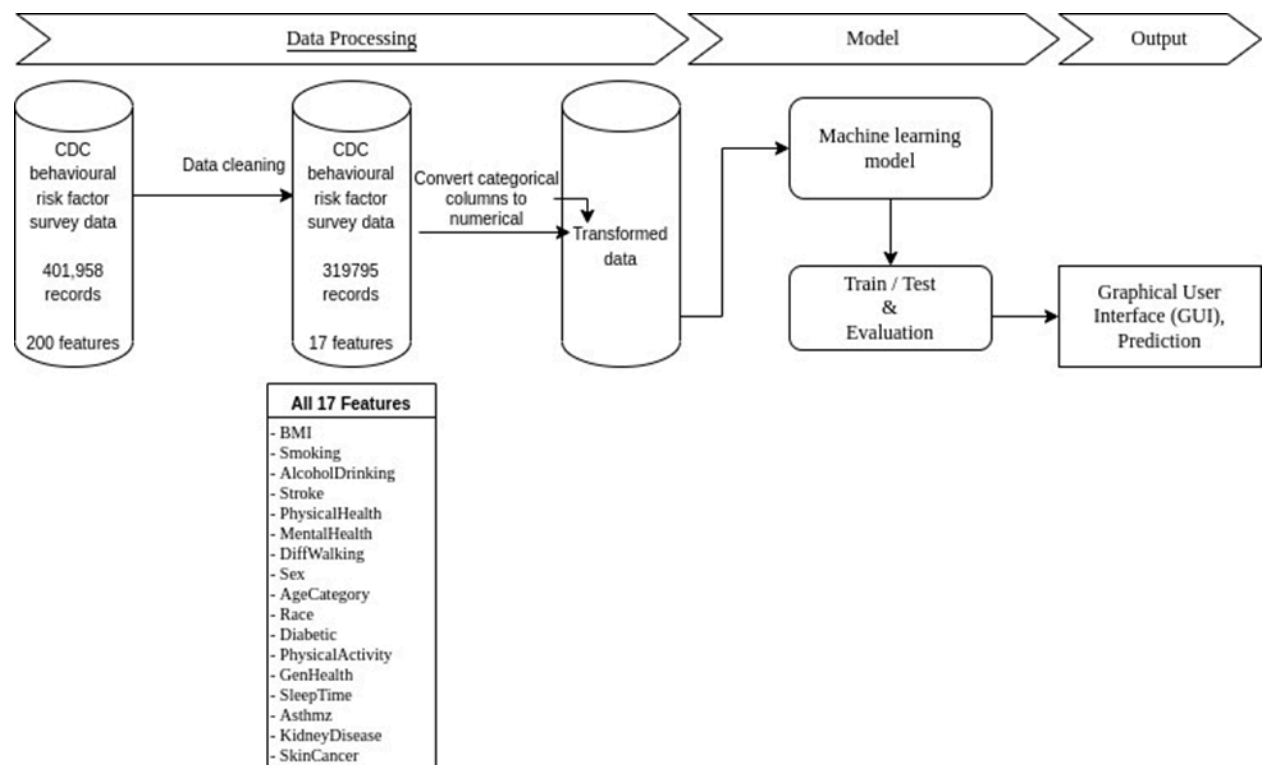


Figure 3.11: System Architecture of the Proposed System

Figure 3.11 shows a high level architectural view of the system. I personally created this diagram to show the architecture of the system.

3.6 FUNCTIONALITY OF THE SYSTEM

The interactions between the system and the user are graphically depicted in a use case diagram. Use case diagrams are important in system design because they serve as a roadmap for designing the system's structure, as well as defining who will use the system and how the user expects to interact with it. The use case diagram for the system is shown below.

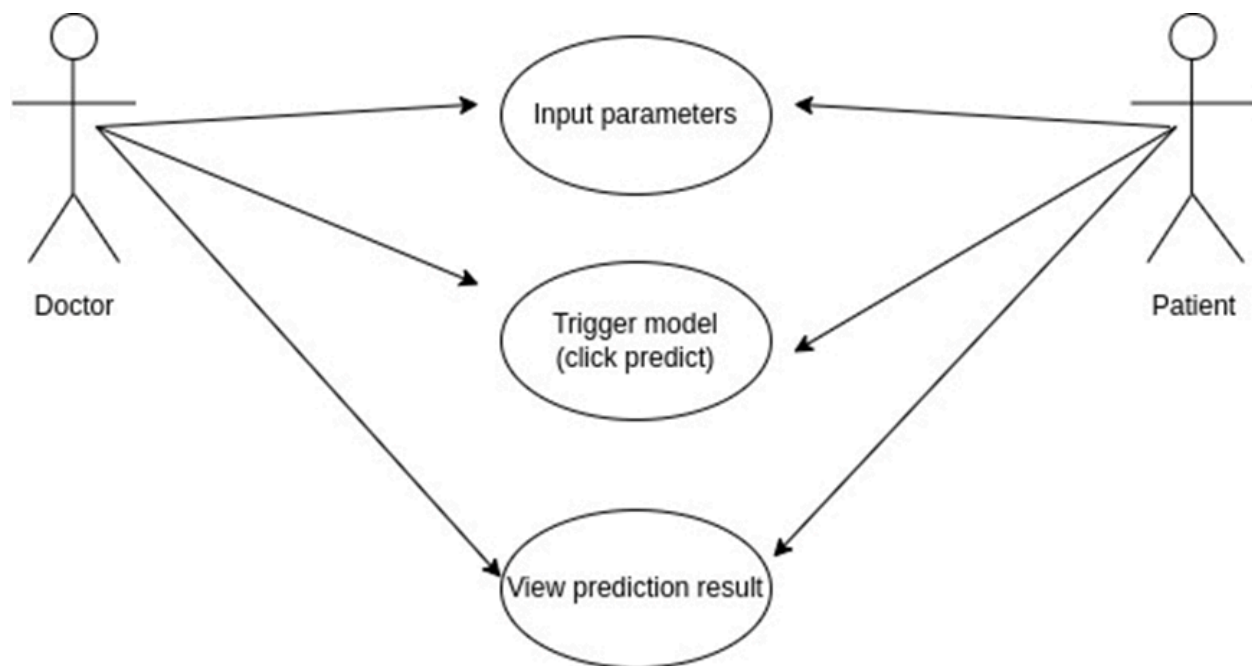


Figure 3.12: Use Case Diagram of the system.

The figure above is the use case diagram depicting the order the users (doctor and patient) can access the system.

CHAPTER FOUR

PROGRAMMING, IMPLEMENTATION AND DOCUMENTATION

4.1 INTRODUCTION TO PROGRAMMING CONCEPTS

The process of writing an executable computer program to fulfil a specific computational objective is known as programming. Analysis, resource consumption, evaluating the accuracy of algorithms, and implementing algorithms in a chosen programming language are at the core of programming (Hartmann, 2013). Language abilities range from basic to advanced. Lower-level languages are more directly tied to the platform on which they are built, whereas higher-level languages abstract the platform to a large part for the programmer.

Web application, programming, and development ideas are used in this work to create an end-to-end heart disease prediction system. The programming of application software that is hosted on external servers and conveyed to a mobile screen through a network is known as web application development. A web page does not need to be downloaded; instead, it is viewed through the internet using web browsers such as Google Chrome, Brave, Safari, and Mozilla Firefox. The vast number of online apps can be created in HTML5, CSS, Python, or JavaScript.

Website development is mainly divided into two aspects, namely, front-end web development and back-end web development.

Front-end web development is the aspect of web development that concerns developing interfaces that users can interact with. It is also referred to as client-side programming. All the visible parts of a website, that is, the client-side, are done using a markup language (e.g. HTML) which describes the structure of the webpage, a stylesheet to add beauty to the website through colors, layout and fonts (e.g. CSS), and a programming language to add functionalities to the website and make it interactive (e.g. Python, JavaScript, PHP).

The back-end is concerned with the invisible part of the website. It is the powerhouse of the front-end. It is used to create the scripts that the web applications use. Scripts could be written in Python, Typescript, JavaScript, and a host of other programming languages. Server-side programming creates an independent interface for the end-user and hides the source code that powers the interface.

4.1.1 Software methodology

This refers to the steps and procedures that must be followed to realise the desired model. Each of the following processes is distinct and crucial for the development of the application, as each state is dependent on the others in some way. The software methodology used for this program is the Iterative Model.

- I. Planning and Requirements
- II. Analysis and Design
- III. Implementation
- IV. Testing
- V. Evaluation

4.2 Stages of Development

Below are the stages for implementing the development model.

I. Planning and Requirements

The first phase, like with other development projects, is to pass through a project initiation stage to plan out the specification report, determine software or hardware needs, and essentially prepare for the future stages of the cycle. A thorough research was done and a document with the resemblance of an SRS (System Requirements and Specification) document was created to serve as a guide.

II. Analysis and Design

Data for this system was gotten from an online data repository, Kaggle. After getting the data, it was cleaned and analysed with pandas, seaborn and other Python libraries. Figure 5 shows the architectural design of the system.

III. Implementation

Machine learning techniques was used in building the heart disease prediction system. Python was used as the primary programming language for building the application, especially the backend part of the software. Steamlit was used for creating the frontend of the application. The machine learning algorithms used are logistic regression, random forest and Support Vector Machine (SVM). However, logistic regression had the highest accuracy, percision, recall and f1-score. It was connected to the streamlit application and used as the main algorithm for predicting heart disease.

IV. Testing

After coding and implementing the current build iteration, the following stage is to go through a set of testing methods to discover and uncover any potential flaws or issues that have arisen. Standard machine learning model evaluation techniques were used to test the application as shown in Figure 4.7.

V. Evaluation

Once all previous steps have been accomplished, it is essential to conduct a detailed assessment of progress up to this point. This allows the user, as well as doctors or any other third parties, to assess where the project is, where it needs to go, what can or should be changed, and so on.

4.2.1 Development Languages Used

This study used Python as the main programming language in the development of the heart disease prediction system. Below is a list of languages and libraries /frameworks that were used in achieving the research objectives.

PYTHON: Python is a general-purpose, high-level, interpreted programming language. Its design concept prioritizes code readability by employing heavy indentation. Python is garbage-collected and dynamically typed.

STREAMLIT: Streamlit is a free, open-source, all-Python framework that allows data scientists to easily construct interactive visualizations and machine learning web apps with no prior knowledge in front-end web development.

4.2.2 Design Algorithm

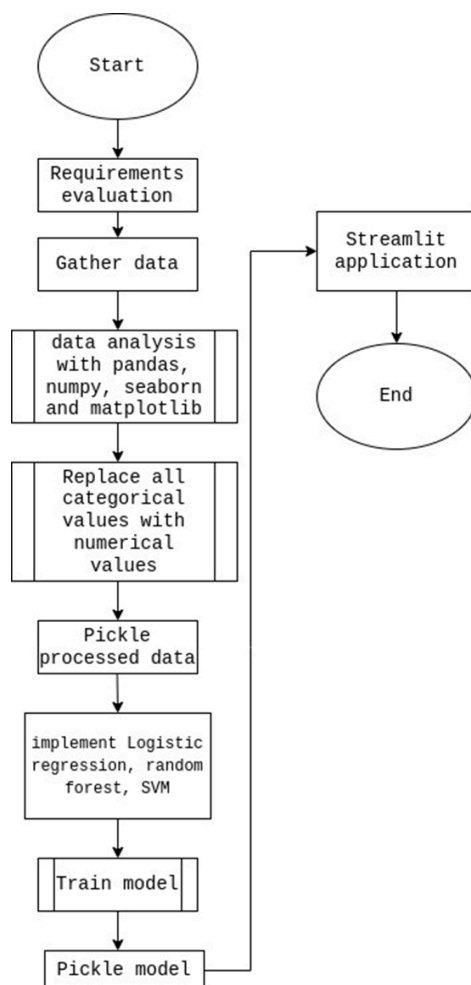


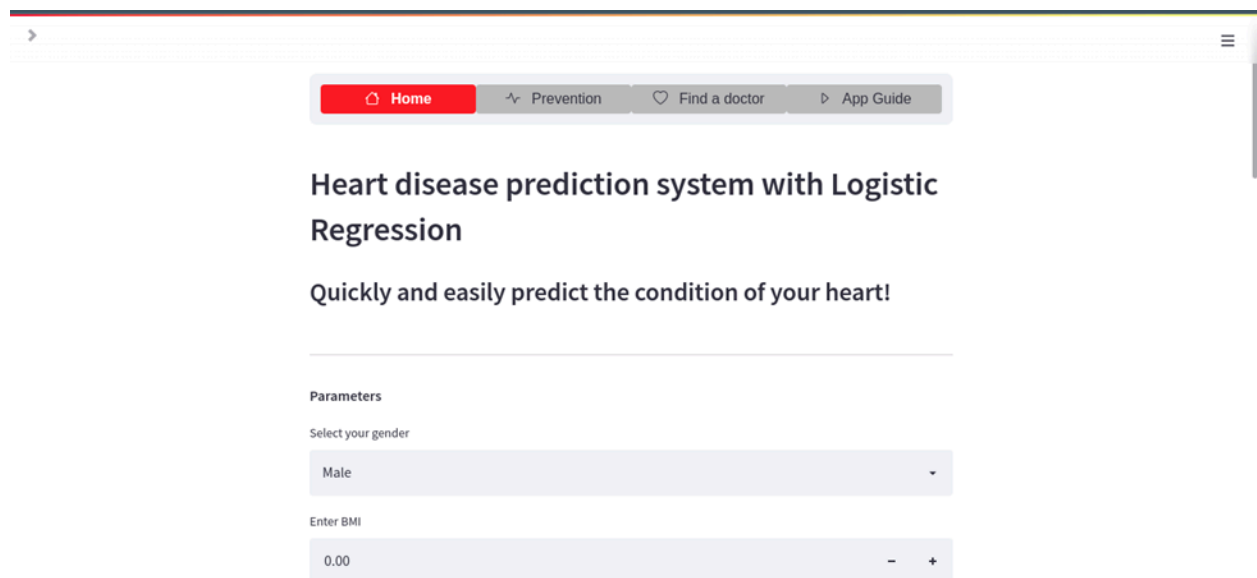
Figure 4.1: Design Algorithm of the Proposed System

The design algorithm above shows the steps taken while designing this system.

An algorithm shows the steps taken to achieve a goal. The design algorithm of this study shows the steps taken to build a heart disease prediction system.

4.2.3 Program Design

In this study, a heart disease prediction system was designed. The system has five screens. It has a home page where users can input parameters and view the prediction results. It has a page where users can read about preventive measures against heart disease, a side panel with information about the application, an embedded page where users can find doctors if the prediction shows they heart disease and an app guide page. The input screen and the output screen are shown in Figure 3.5 above.



The screenshot displays the home screen of a web application titled "Heart disease prediction system with Logistic Regression". At the top, there is a navigation bar with four buttons: "Home" (highlighted in red), "Prevention", "Find a doctor", and "App Guide". Below the navigation bar, the main heading reads "Heart disease prediction system with Logistic Regression", followed by the tagline "Quickly and easily predict the condition of your heart!". Underneath, there is a section titled "Parameters" containing two input fields: "Select your gender" with a dropdown menu currently showing "Male", and "Enter BMI" with a text input field showing "0.00" and increment/decrement buttons.

Figure 4.2: Home Screen of the Application

This is the screen the user will see first when the web page is loaded.

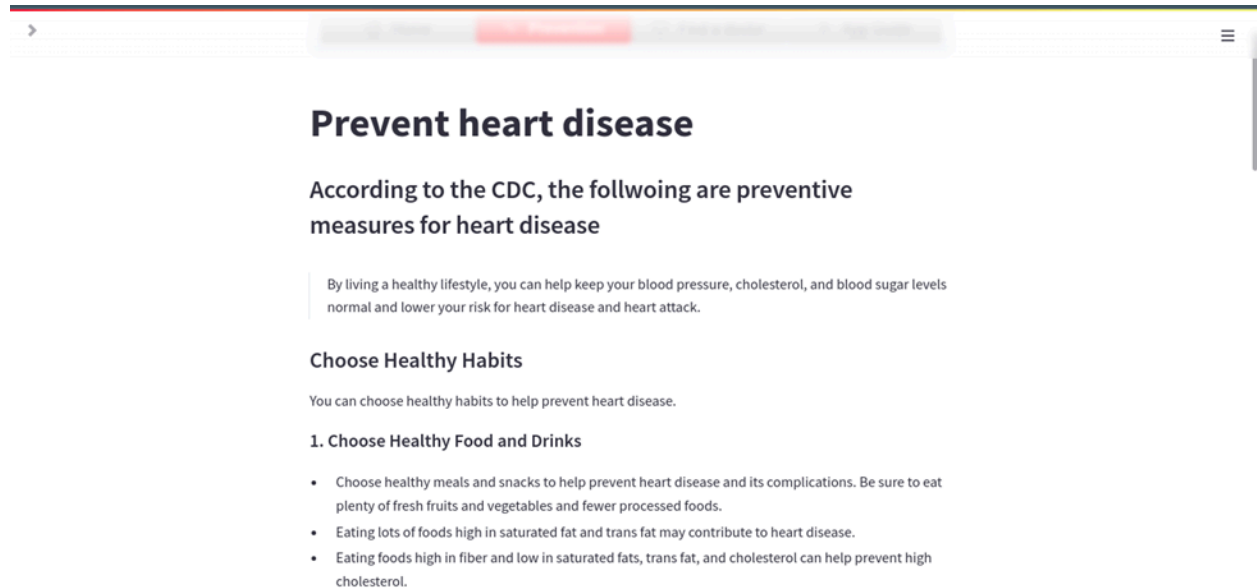


Figure 4.3: Preventive Measures Screen

The figure shows the preventive measures against heart disease.

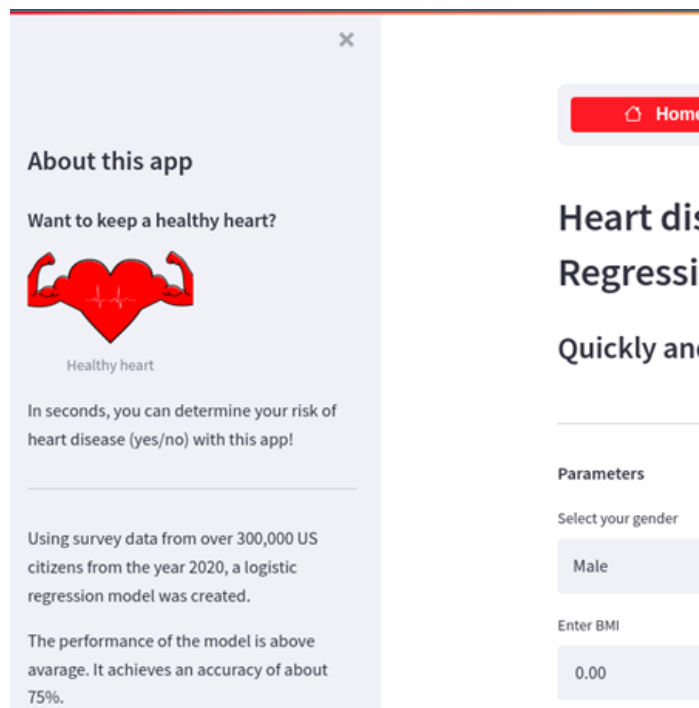


Figure 4.4: About Screen Showing Information About the System

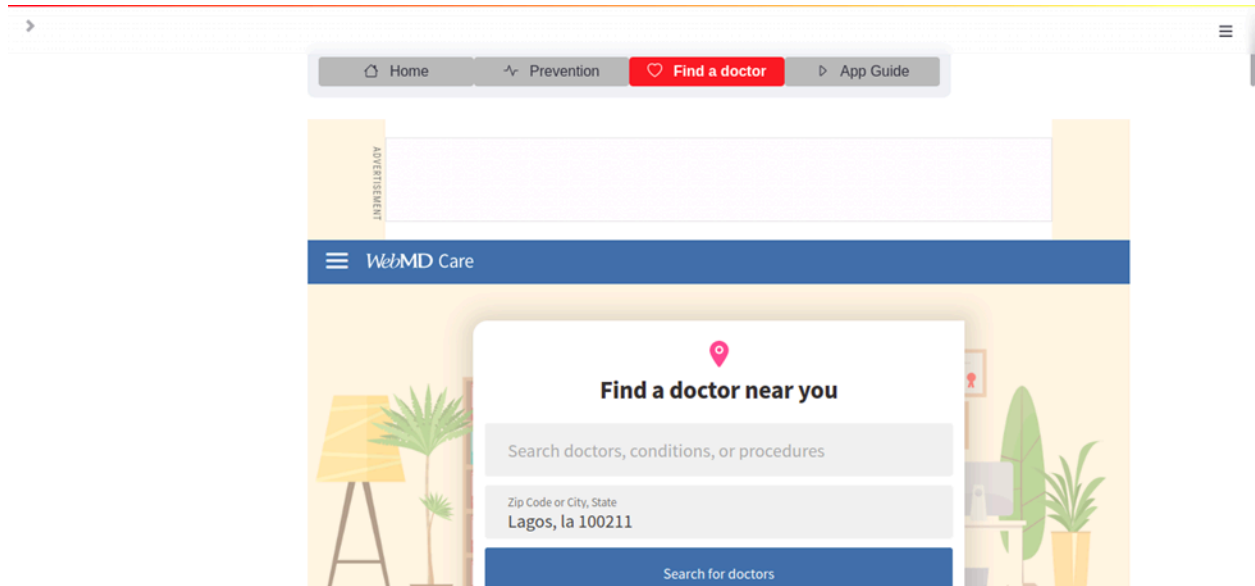


Figure 4.5: Find a Doctor Screen

This embedded page allows patients to find doctors near them.

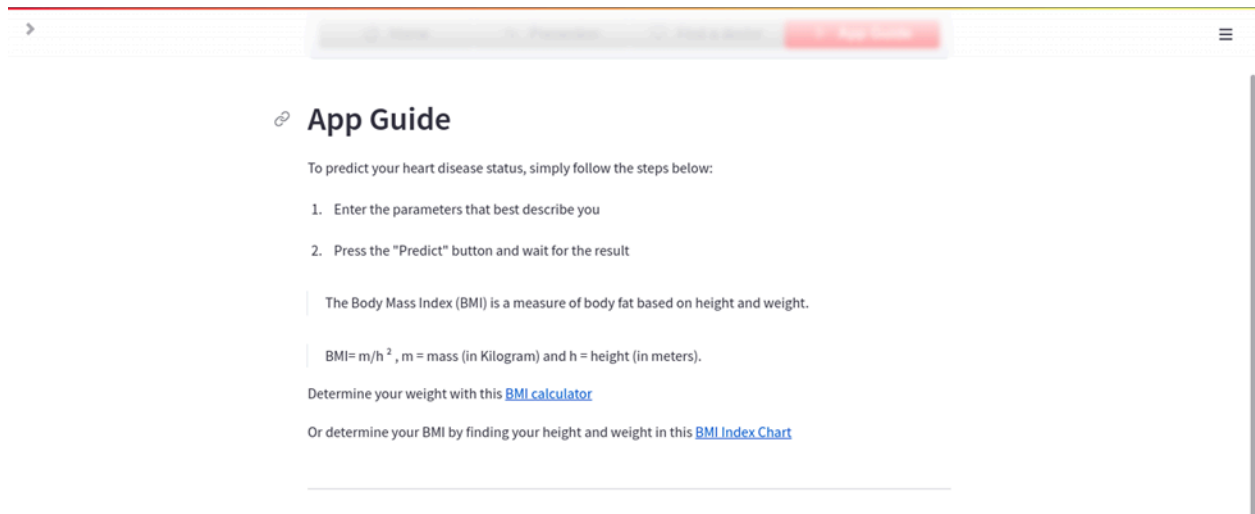


Figure 4.6: App Guide Screen

This screen shows information on how to use the system.

4.2.4 DEFINITION OF PROGRAM VARIABLES

The program variables used in this study are described below:

- i. BMI – Body Mass Index of patient.
- ii. Smoking – Assesses if the patient has smoked at least 100 cigarets in their entire life.
- iii. AlcoholDrinking – Assesses if the patient has had more than 14 drinks of alcohol (men) or more than 7 (women) in a week.
- iv. Stroke – Assesses if the patient has a stroke.
- v. PhysicalHealth – Assesses the patients' physical health in the last 30 days.
- vi. MentalHealth – Assesses the patients' mental health in the last 30 days.
- vii. DiffWalking – Checks if the patient has difficulty in walking.
- viii. Sex – represents the gender of the patient.
- ix. AgeCategory – Represents the age category of the patient (18 and above).
- x. Race – Represents the race of the patient.
- xi. Diabetic – Assesses if the patient has diabetes.
- xii. PhysicalActivity – Assesses if the patient has been physically active.
- xiii. GenHealth – Assesses the general health of the patient (Excellent, very good, good, fair, poor).
- xiv. SleepTime – Represents the average sleep time of the patient.
- xv. Asthma – Assesses if the patient has asthma.
- xvi. KidneyDisease – Assesses if the patient has kidney disease.
- xvii. SkinCancer – Assesses if the patient has skin cancer.

4.2.5 JUSTIFICATION OF THE PROGRAMMING LANGUAGE USED

Python is among the most understandable computer languages that exist due to its streamlined syntactic structure and lack of complexity, which favours human language. Python scripts may be produced and performed significantly quicker than all other programming languages due to their ease of understanding and use.

Furthermore, Python has a rich library that includes foundational objects, which eliminates the need for programmers to build software from the ground up each time. Continuous processing of data is required for machine learning applications, and the libraries provided by Python enable developers to process, access, and change data. These libraries are among the most comprehensive machine learning and artificial intelligence libraries available.

4.3 SYSTEM TESTING

System testing, also known as system evaluation is a process in which a quality assurance team assesses how distinct system components communicate with one another. System testing confirms the efficacy of an application. This procedure, which is a sort of black box testing, is concerned with an application's usability. For example, system testing may examine whether all user inputs result in the desired outcome across the software.

Model validation approaches were used to evaluate the model's performance, which was then compared to the objectives of the study. Following a comprehensive assessment, the system functions exactly as anticipated. It can determine whether or not a patient has cardiac disease and then translate that information into a probability.

Assessing machine learning model differs greatly from evaluating traditional software. Depending on the classifier used, there are basic libraries from which measures may be imported to validate the model's performance. Model assessment is the method in question. Another curve is the AUC-ROC (Area Under the Curve Receiver Operating Characteristic) curve. This curve

displays the model's data classification score. The system designed was evaluated using the confusion matrix, classification report, and AUC-ROC curve.

The data was first separated into two sets: test and train. The model was validated using the test set. The system was tested using the accuracy score, precision, f1-score, recall, confusion matrix, and an AUC-ROC curve. The AUC-ROC curve indicates how accurate the system is in predicting whether or not a patient has heart disease. An AUC value of 1.0 indicates that the model can determine whether a patient has heart disease or not. This system has an AUC of 0.841.

Confusion matrix

```
[[4073 1410]
 [1192 4275]]
```

Classification report

	precision	recall	f1-score	support
0	0.74	0.77	0.76	5265
1	0.78	0.75	0.77	5685
accuracy			0.76	10950
macro avg	0.76	0.76	0.76	10950
weighted avg	0.76	0.76	0.76	10950

Figure 4.7: System Testing Results

4.4 OPERATIONAL PROCEDURE

The design of the system is clean and easy to use. Special training is not needed to use the system. Also, no login or signup is needed. The operational procedure is as follows:

- I. A user first opens the web page from their device with internet connection.
- II. After the page has loaded, the user has to click on the 'App guide' to read how to use the application.
- III. The user then goes back to the 'Home' page and scrolls down to fill in the parameters as described in each textbox.
- IV. After filling in the appropriate details, the user must click on the "predict" button to view the result of the system.
- V. If the prediction is below 50%, the user can go the prevention page to view the measures to prevent heart disease.
- VI. If the prediction is 50% and above, the user can go to the 'find a doctor' page to search for a doctor. Early treatment is key.

4.4.1 Hardware Requirements

The hardware requirements for this system are listed below:

- I. A minimum of 450GB hard disk drive.
- II. A minimum of Core i5 1.6 GHz Intel Processor.
- III. A minimum of 4GB RAM.
- IV. A CD-ROM drive.
- V. An SVGA monitor.
- VI. Input device (Keyboard, mouse, etc.).

4.4.2 Software Requirements

The software requirements for this system are listed below:

- I. Any Linux distribution i.e. Ubuntu, Arch Linux, Fedora, Linux Mint, CentOS.
- II. Windows 10 and above.
- III. Python 3.7 and above.
- IV. Virtual environment.
- V. Web browsers.
- VI. Terminal.

4.4.3 Running and Accessing the Program

The following procedure shows how to run the program:

- I. Activate the virtual environment in the root folder/folder where it was installed with
`source .newvenv/bin/activate`.
- II. Cd into the project folder.
- III. Start project with Streamlit run app.py
- IV. The web page will automatically open up.

CHAPTER 5

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 SUMMARY

Machine learning is an emerging and evolving subfield of Artificial Intelligence (AI) with the ability to analyse data and make accurate predictions from the data. Artificial intelligence has gained traction in healthcare and machine learning has been used to implement several solutions to automate previously manual processes in the healthcare sector. Heart disease is a major and silent killer disease today. Treating it in its early stages is essential. With machine learning algorithms, doctors can know how susceptible their patients are to heart disease, and patients can know how susceptible they are to heart disease. This study discussed a heart disease prediction system with the details of the lifestyle of patients as the input for the system to predict their probability of having heart disease. The system performed well above average, with an accuracy of 76%. Therefore, it can be relied upon to an extent. With some enhancements, doctors and patients can fully utilise the system.

5.2 CONCLUSION

Heart disease is among the list of the most dangerous illnesses in the world. It is essential to create a method to precisely and effectively forecast heart disease in people given the rise in fatalities caused by cardiac disease. The system developed serves as a good auxiliary for predicting the chance of heart disease in patients. It reduces the delays associated with the existing system because carrying out lab tests, performing ECGs, and performing other tests can be time-and effort-consuming. This system, was built to act as a forerunner to medical diagnosis, leading to an early awareness of the other side of people and possible lifestyle adjustments. Logistic regression, was used to classify patients with heart disease and those without heart disease. Further steps were taken to determine the probability of the detection. In the future, the system can be enhanced with quality data and more features can be added to extend its functionality.

5.3 RECOMMENDATIONS

Cardiologists should examine the possibility of having a pre-diagnosis before an actual diagnosis so that they do not spend time carrying out blood tests and ECGs for a person that may not have a heart disease. Individuals should have this system at their disposal to frequently evaluate their chances of having a heart disease so that they can be mindful of their way of living.

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